SQL Hospital Data Analysis

Question 1

Total Number of Patients

Write an SQL query to find the total number of patients across all hospitals.

1 SQL QUERY

```
SELECT SUM (patients_count) AS Total_Patients
FROM Hospital_data;
```

Result :

Total_patients
9347.

Question 2

Average Number of Doctors per Hospital Retrieve the average count of Doctors available in each hospital.

2 SQL QUERY

```
SELECT avg (doctors_count) AS avg_doctors
FROM hospital_data;
```

Result :

AVG_doctors	26.71000000000000000

Question 3

Top 3 Departments with the Highest Number Patients

Find the top 3 hospital departments that have the highest number of patients.

3 SQL QUERY

```
SELECT department ,SUM (Patients_count) AS total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients DESC
 LIMIT 3;
```

Result :

Total Rows:100

Question 4

Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.

4 SQL QUERY

```
SELECT Hospital_name, medical_expenses
FROM hospital_data
ORDER BY medical_expenses DESC
LIMIT 1;
```

Result :

Hospital_name	Medical_expenses
Healing Touch	49955.41

Question 5

Daily Average Medical Expen

Calculate the average medical expenses per day for each hospital.

5 SQL QUERY

```
SELECT medical_expenses hospital_name, /
 (discharge_date - admission_date) AS expenses_per_day
FROM hospital_data;
```

Result :

Hospital_name	Expenses_per_day
Welleness cliniv	31364.880000000000000
Fortis Care	4298.19990909090909091

Question 6

Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

6 SQL QUERY

```
SELECT hospital_name,
(discharge_date - admission_date)
AS stay_days
FROM hospital_data
ORDER BY stay_days DESC
LIMIT 1;
```

Result :

Total rows: 1

Hospital_name	Stay_days
Apollo Health	15

Question 7

Total Patients Treated Per City

Count the total number of patients treated in each city.

7 SQL QUERY

```
SELECT Location, SUM (patients_count) AS total_patients
FROM hospital_data
GROUP BY Location;
```

Result :

Total roew :10

Location	Total_patients
Bangalore	955
Jaipur	1505
Lucknow	1264

Question 8

Average Length of Stay Per Department

Calculate the average number of days patients spend in each department.

8 SQL QUERY

```
SELECT department,
AVG (discharge_date - admission_date) AS
avg_stay_days
FROM hospital_data
GROUP BY department;
```

Result :

Total rows:10

Department	Avg_stay_days
Dermatology	3.600000000000000000

Oncology	8.111111111111111111
ENT	8.083333333333333333

Question 9

Identify the Department with the Lowest Number of Patients

Find the department with the least number of patients.

9 SQL QUERY

```
SELECT department,
Sum (patients_count) AS Total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

Result: :

TOTAL ROWS:1

Department	Total_patients
Cardiology	544

Question 10

Monthly Medical Expenses Report

Group the data by month and calculate the total medical expenses for each month.

10 SQL QUERY

```
SELECT extract (month from admission_date) AS month,
SUM (medical_expenses) AS
Total_expenses
FROM hospital_data
GROUP BY month
ORDER BY month;
```

Result: :

Total rows 12

Month	total_expenses
1	173971.54
2	301722.72
3	199247.42
4	88995.93